Arush Vyas

Boston, MA | arushv16@gmail.com | linkedin.com/in/arush-vyas | github.com/arushv1

Education

Boston University — Boston, MA
Bachelor of Science in Biomedical Engineering | ***GPA**: 3.83* — *Expected Grad: May 2027*

Technical Skills

Programming: Python, PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn, MLflow, R
Medical Imaging: QuPath, ImageJ, 3D Slicer, OpenSlide, wsInfer, Biopython, Groovy, DICOM, MONAI, CTA Image Analysis
Infrastructure: Linux, Git (GitLab, GitHub), AWS (EC2, S3, EKS, SageMaker), Docker, Bash, PowerShell, HPC (BU SCC)

Relevant Experience

Health AI Engineering Intern — May 2026 – Present
Boston Scientific Corporation — *Marlborough, MA*

- Developed and optimized an internal 3D CTA segmentation pipeline for a vascular-therapies AI medical-imaging product, accurately distinguishing artery classes across scans and achieving a mean artery Dice score of 0.87 and clDice of 0.94
- Benchmarked 10+ model variants across different loss functions, attention mechanisms, and foundation-model fine-tuning, identifying vessel-specific objectives and attention gates as the key drivers of a +0.22 Dice gain over baseline model
- Co-invented patent-pending medical-imaging technology, translating clinical requirements into an AI-driven technical solution
- Presented key technical findings, model performance, and product impact to directors and executive leadership

Undergraduate Deep Learning Researcher — Sep. 2025 – Present
Tian Lab | Computational Imaging Systems — *Boston, MA*

- Curating the largest annotated dataset of CTE and Alzheimer's brain tissue histopathology at BU, with 2,500+ annotated tissue patches from 40+ whole slide images (WSIs) and built a diagnosis manifest of 760+ WSIs (control, RHI, CTE, AD)
- Streamlined WSI data collection to minutes per slide by introducing QuPath's multi-resolution viewing and Groovy-scripted tile extraction to the lab's pipeline, enabling scalable dataset curation of vascular features used for downstream model development

Lab Assistant — Sep. 2025 – Present
Boston University's Bioengineering Technology & Entrepreneurship Center — *Boston, MA*

- Implement and maintain a RAG-based knowledge base encompassing 1,800+ BTEC documents across 12 subject areas
- Prepare BTEC resources for AI retrieval by cleaning, restructuring, and converting files into consistent, model-ready formats
- Benchmarked different LLMs (GPT 5.4/4.1) on answer accuracy using a test set of 30 questions drawn from real assistant queries

Projects

VascuPath | *QuPath, Python, PyTorch, Groovy, WsInfer, OpenCV, OpenSlide, MLflow* — Jan. 2026 – Present

- Automating what currently takes neuropathologists hours of manual analysis: an AI pipeline that segments large-scale whole slide images to detect and quantify vascular changes in CTE and Alzheimer's, handling gigapixel-level brain slides
- Replaced three separately trained networks with one multi-head model built on a neuropathology ViT-L foundation backbone, achieving 99.6% stain classification and 96.7% vessel detection accuracy on 500 test patches from Brightfield H&E slides
- Reduced per-slide inference time by 70% (from 5–6 to 1–2 minutes) through a shared embedding feeding stain and vessel prediction heads, scaling processing to hundreds of slides on BU's Shared Computing Cluster (SCC) for feature extraction
- Developed a weakly supervised attention-based MIL model that reached held-out test AUCs above 0.90 for distinguishing controls from repetitive head impact (0.92), CTE (0.91), and high-stage AD (0.94)

Threshold | *Python, TypeScript, Next.js, React, Claude API, pytest, Hypothesis, Vercel* — Oct. 2026

- Built an eligibility linter and patient trial matcher covering 1,011 recruiting Massachusetts cancer trials, combining Clinical Trials.gov, CDC PLACES and MassGIS environmental-justice data to show which communities restrictive criteria exclude
- Engineered an LLM extraction pipeline (Claude, forced tool-use schema over 13 categories) that extracted 3,839 restrictions, all verified word for word against the source text, reaching 0.97 precision and 0.94 F1 (0.95 held-out) on 42 human-labeled trials)

Publications & Awards

- Vyas, A., Bansal, K., et al. "Optimizing Non-Invasive Brain-Computer Interfaces for Motor Rehabilitation in Individuals with Spinal Cord Injury." STEM Students on the Rise, Rice University, Oct. 2025. https://dx.doi.org/10.2139/ssrn.5605290
- 2nd Place in Health and Biotech Hack, Boston University's DS + X Hackathon, Oct 2026
- UROP Symposium Convergent Research Poster Award, Health Across the Lifespan, 1 of 4 inaugural award winners, 2026
- 2x Student Research Award Recipient, Boston University's Undergraduate Research Opportunities Program (UROP), 2025, 2026
- Harold C. Case Merit Scholarship Recipient for Academic Excellence, Boston University, 2026, one of 15 selected recipients
- 3rd Place in Early-Stage Lung Cancer Phenotyping track, MedAI Hackathon, Boston University, April 2026